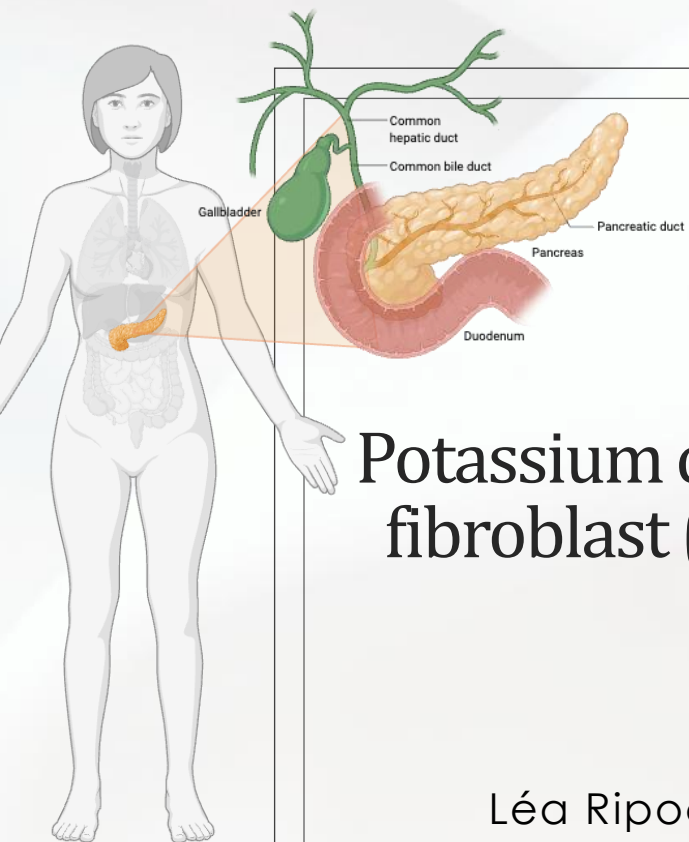
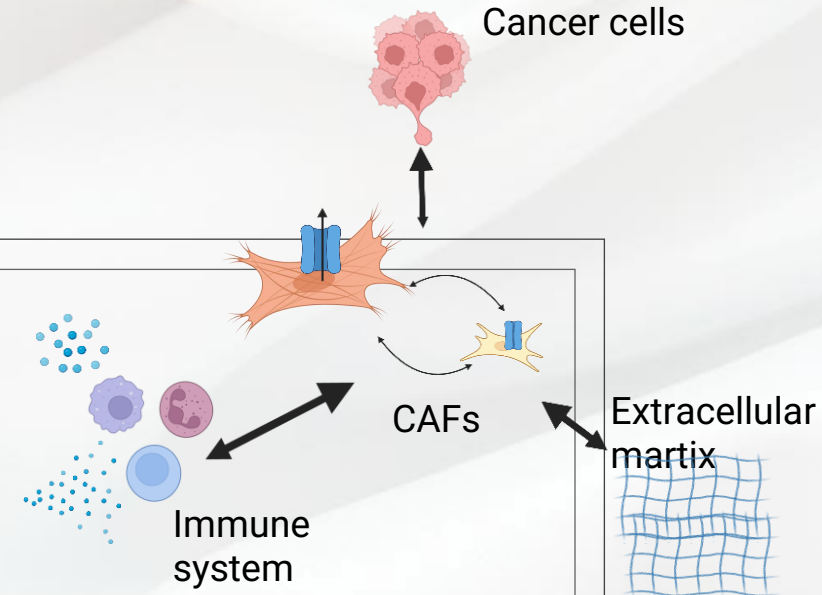


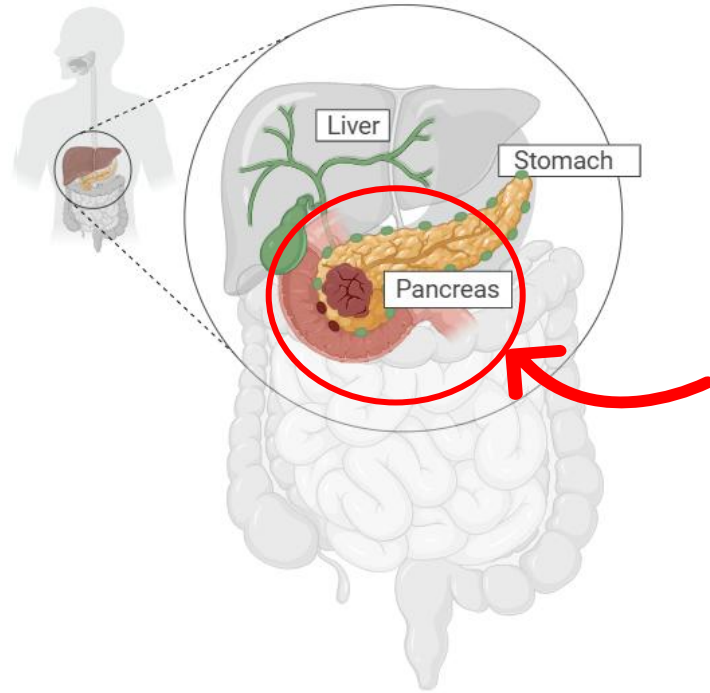


Potassium channel BKCa as a key regulator of cancer associated fibroblast (CAF) in pancreatic ductal adenocarcinoma (PDAC)

Léa Ripoché, iBV, team « ion channels and pathologies » (O. SORIANI)



Pancreatic Ductal adenocarcinoma (PDAC) : a major clinical challenge



Function :

- Secretion of pancreatic juices (exocrine function)
- Insulin secretion --> blood sugar regulation (endocrine function)

PDAC

- 5-year survival less than 13%
- Predicted as 2nd cancer-related death in 2030

Late diagnosis

Dense and pro-tumorigenic
microenvironment

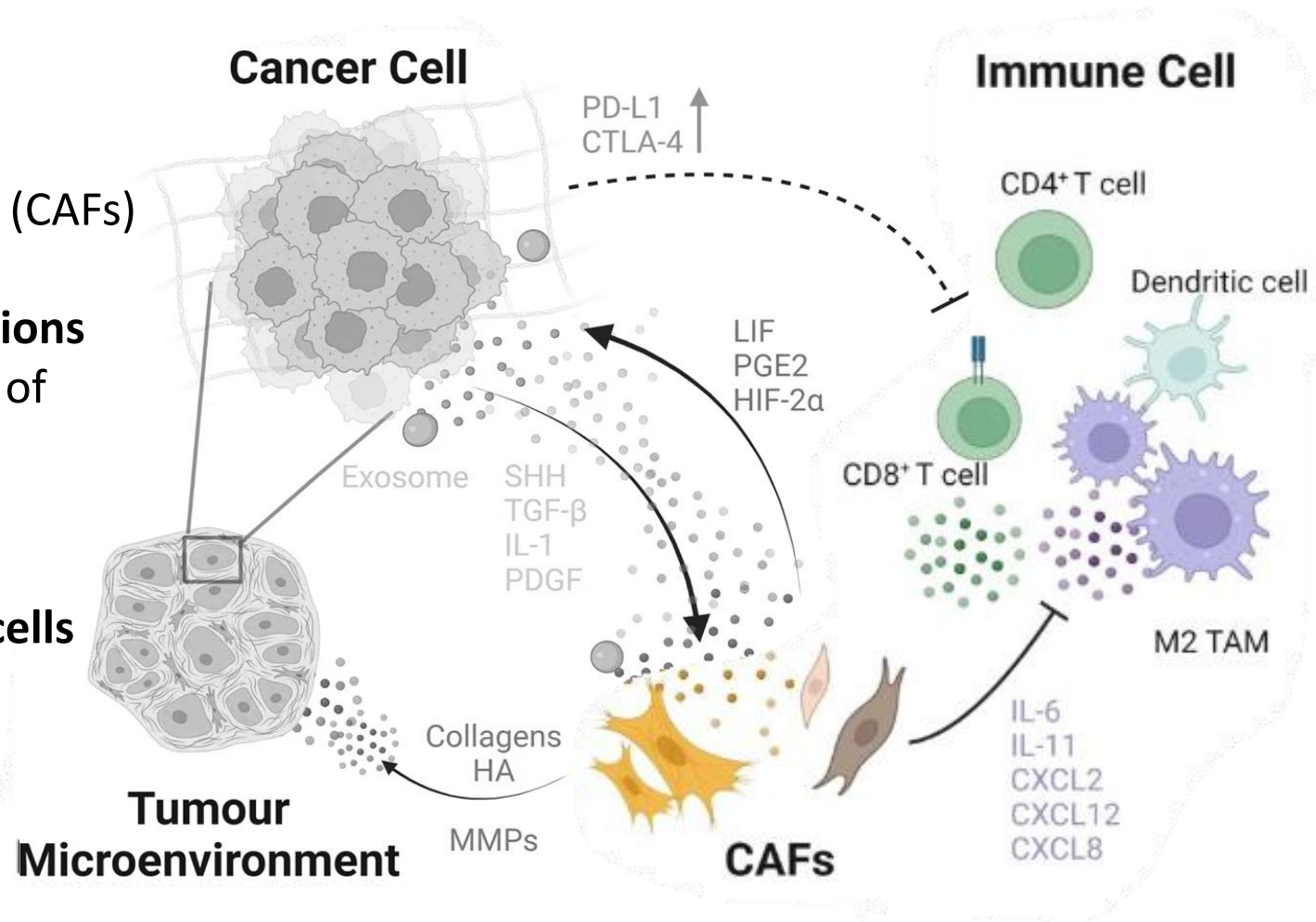
Tumor microenvironment : a crosstalk between cell compartments

- 1 Cancer cells :**
 - Activate fibroblasts.

=> Cancer Associated Fibroblasts (CAFs)
- 2 CAFs :extracellular matrix secretions**

=> dense stroma. Hinders access of immune cells and treatments to cancer cells
- 3 CAFs :interaction with immune cells**

=> immunosuppression



Zang et al., Cell Death and Disease, 2022

Tumor microenvironment : a crosstalk between cell compartments

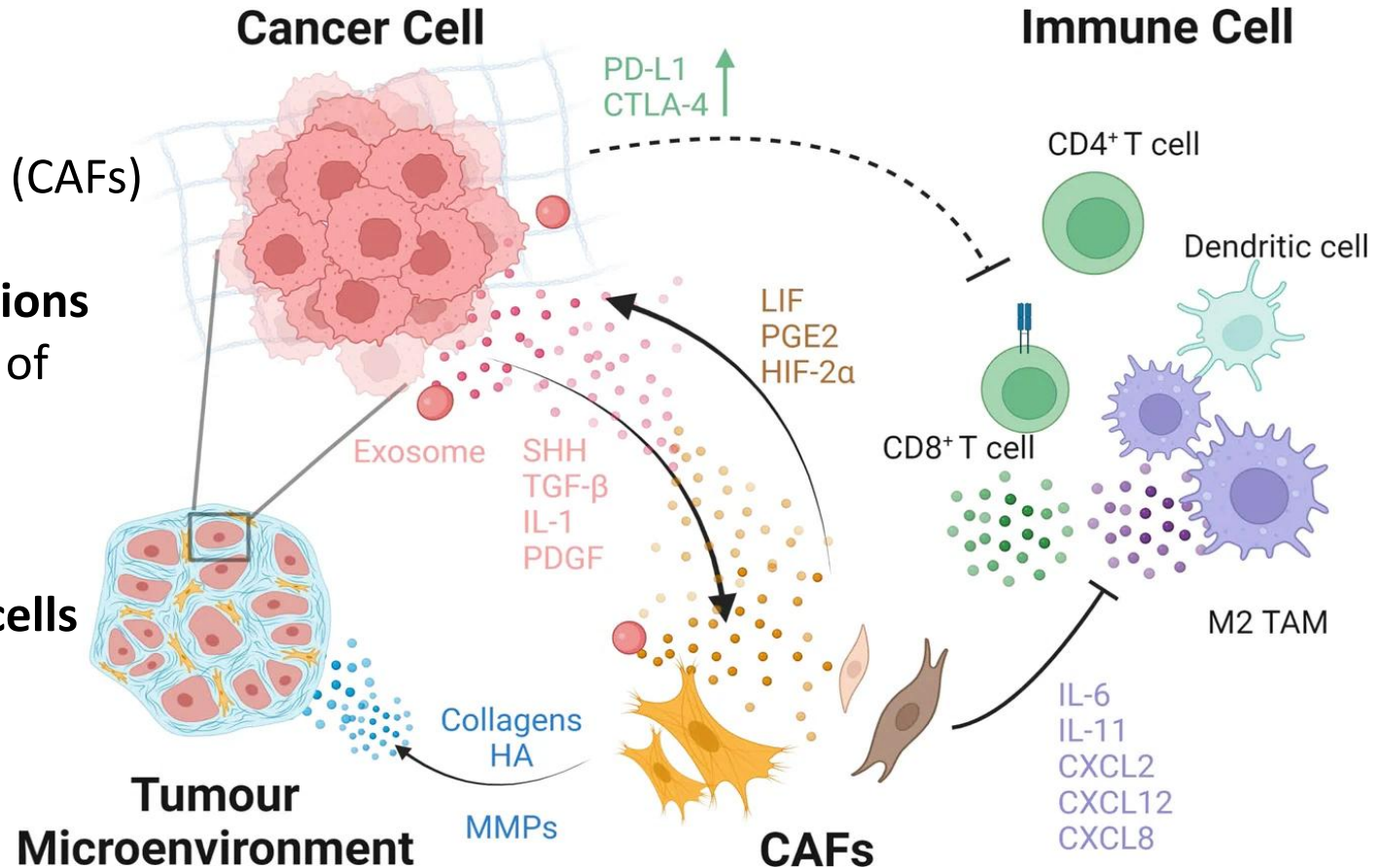
- 1 Cancer cells :**
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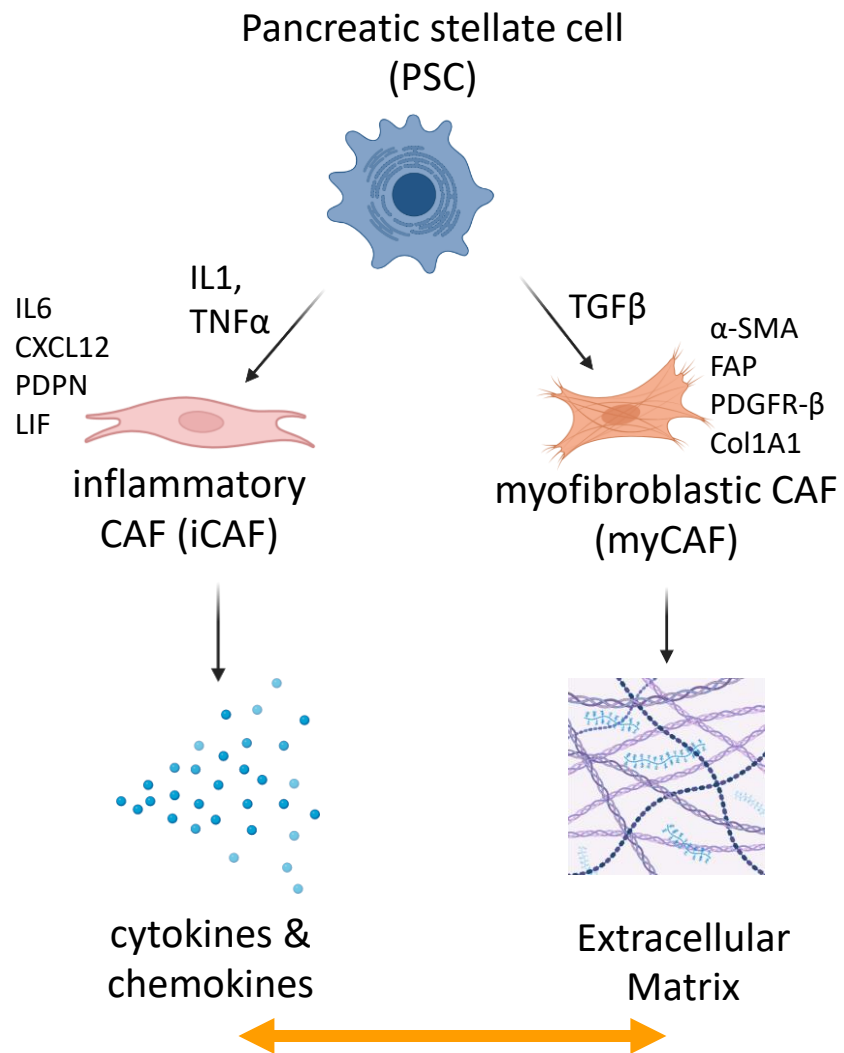
=> immunosuppression

=> pro-tumorigenic Crosstalk



Zang et al., Cell Death and Disease, 2022

CAFs subsets : understanding functional diversity



- Inflammatory CAFs (iCAFs)

Stimulated by IL1, TNF α

--> immune compartment

- Myofibroblastic CAFs (myCAFs)

Stimulated by TGF β

--> Extracellular Matrix



High heterogeneity

Tumor restricting CAFs

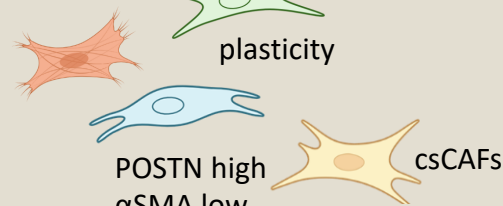


Subset diversity

plasticity

POSTN high
 α SMA low

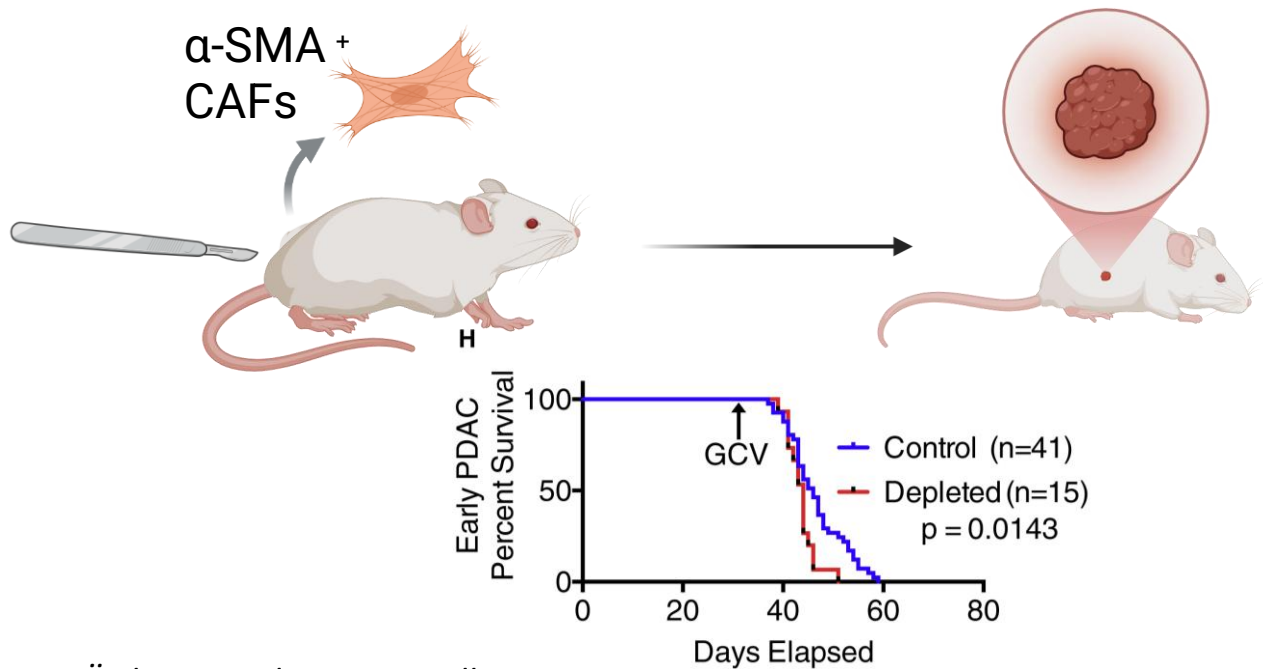
csCAFs



Öhlund et al., J Exp Med, 2017

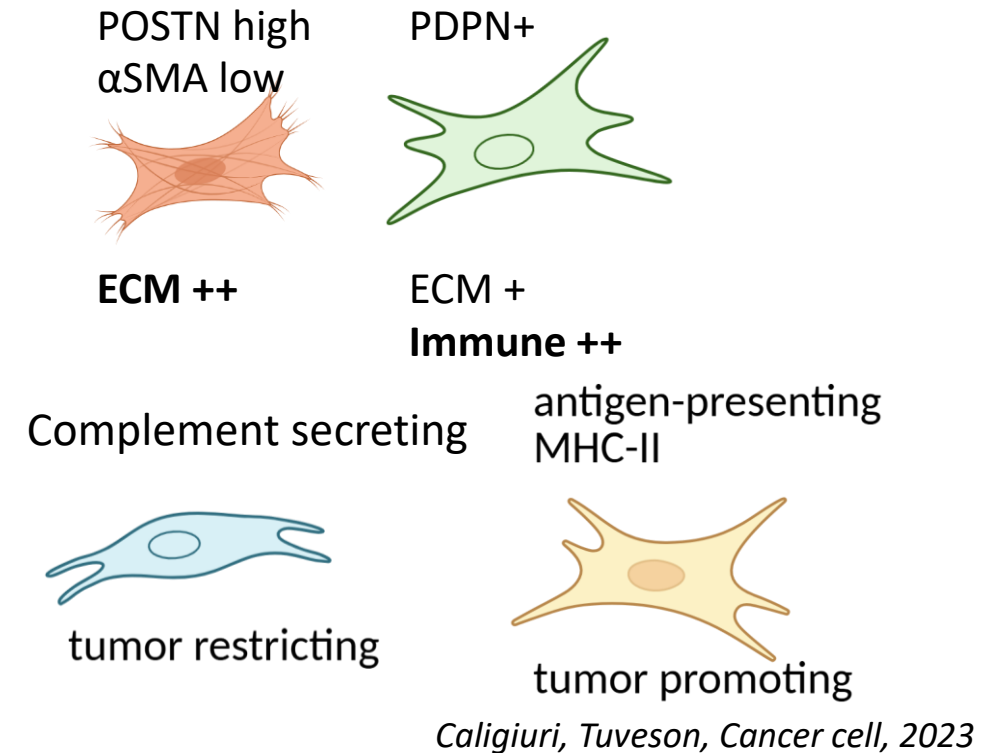
CAFs subtypes : high heterogeneity and plasticity

Depletion of α -SMA⁺ CAFs leads to invasive tumors



Özdemir et al., Cancer Cell, 2014

New emerging subtypes



New paradigm : CAF continuum

myCAFs

iCAFs



CAFs are key therapeutic targets

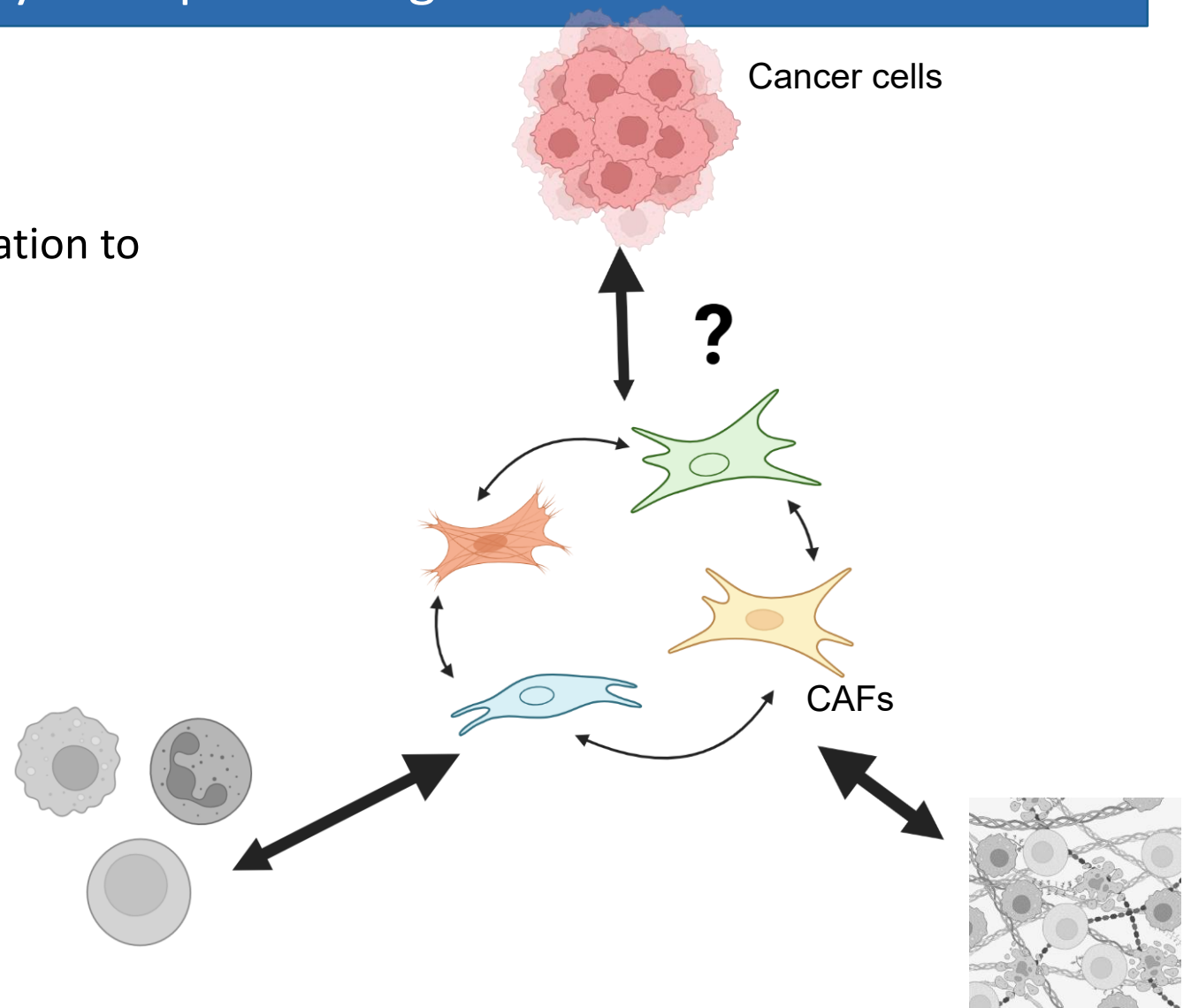
Therapeutic targets :

Trying to control plasticity → reprogramming to quiescent or anti-tumorigenic CAFs

Research strategy

Understanding of CAFs physiology and plasticity

How do they respond to their environment ?



CAFs are key therapeutic targets

Therapeutic targets :

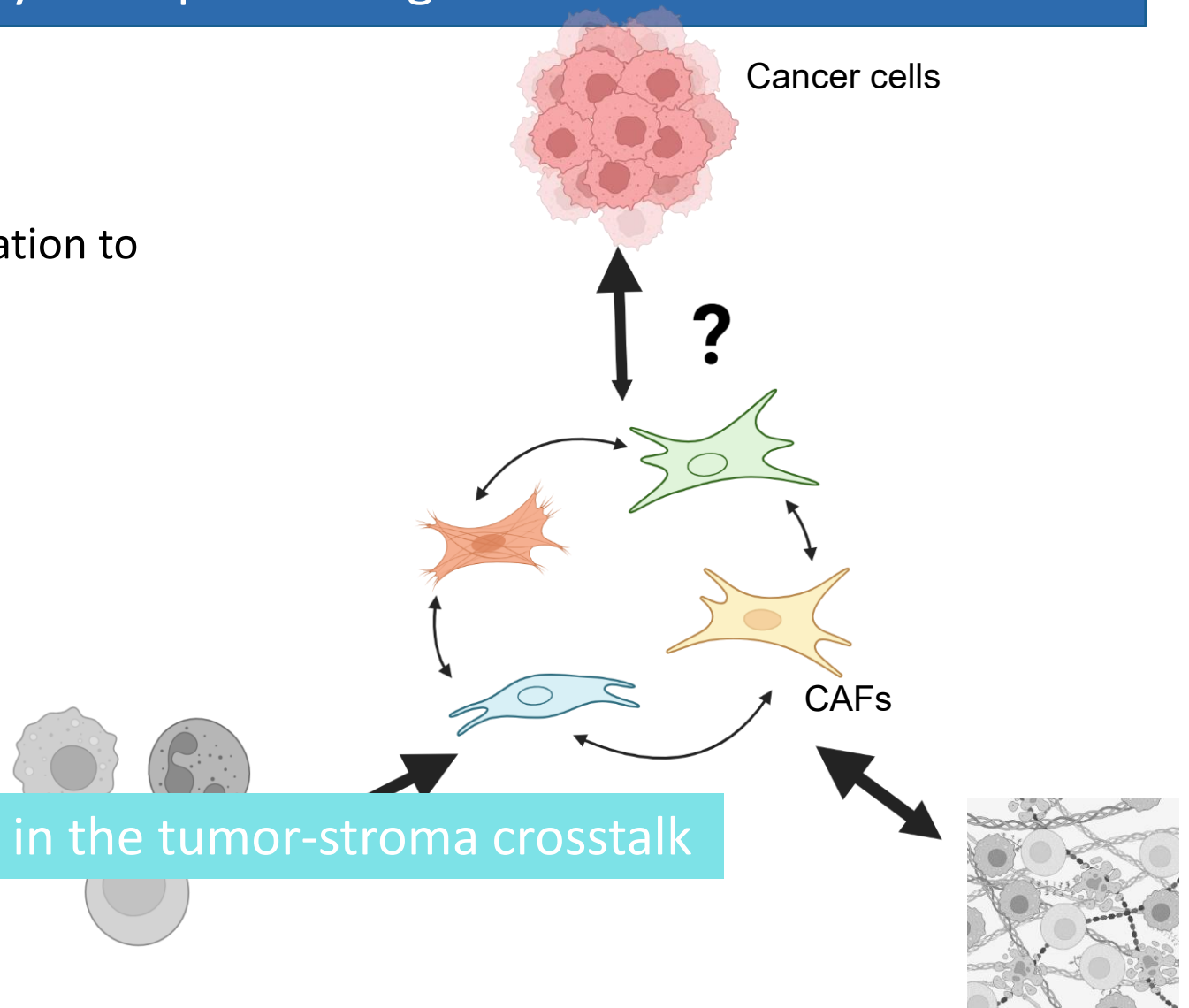
Trying to control plasticity → reprogramming to quiescent or anti-tumorigenic CAFs

Research strategy

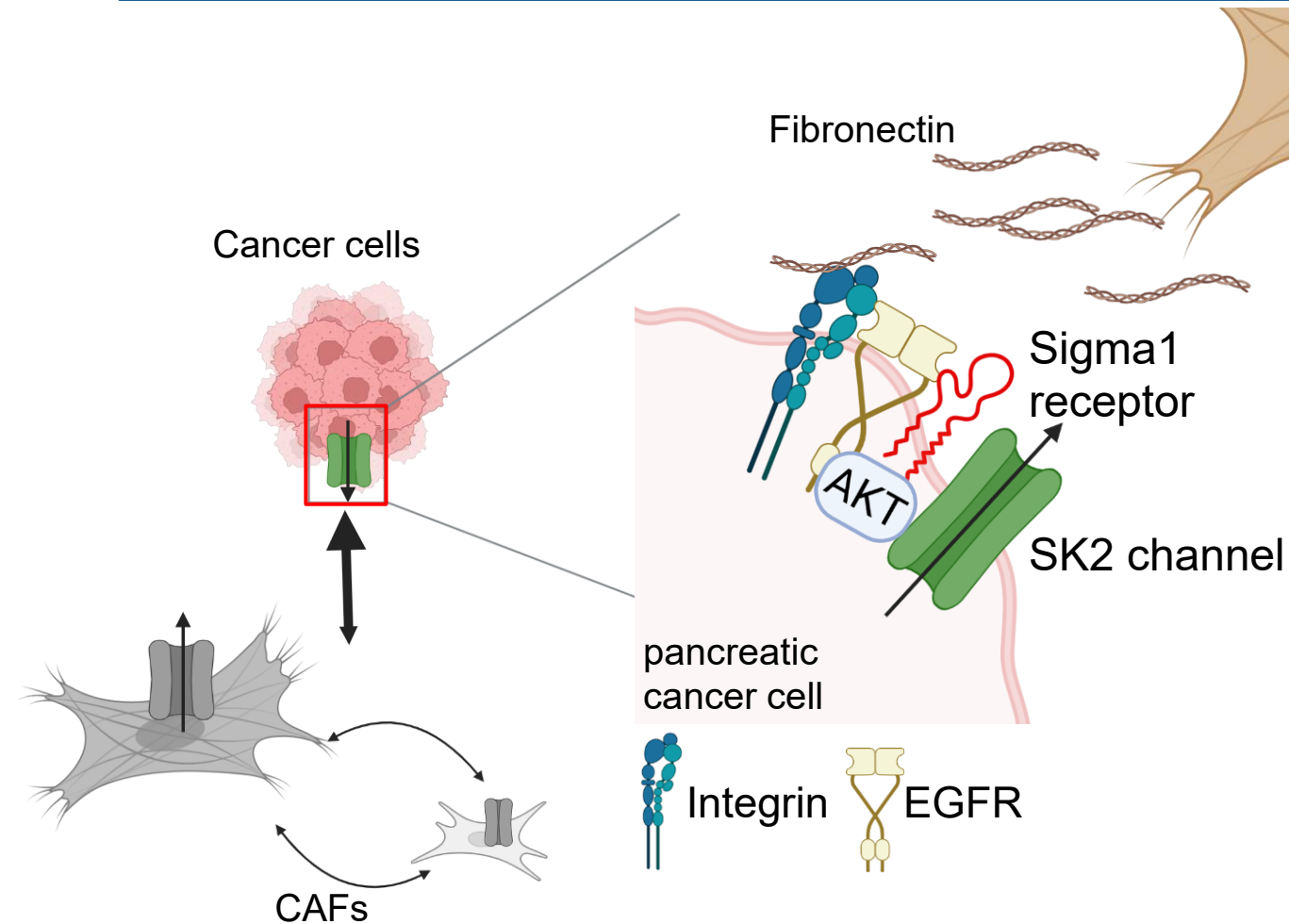
Understanding of CAFs physiology and plasticity

How do they respond to their environment ?

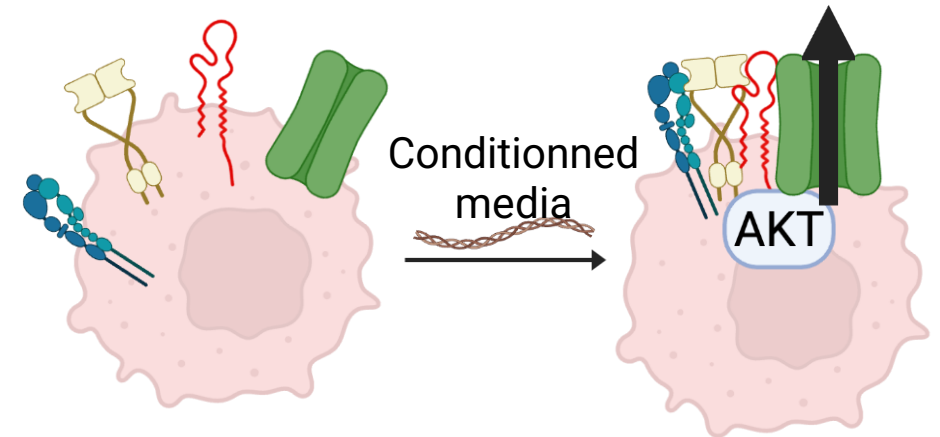
=> Ion channels as key intermediate in the tumor-stroma crosstalk



SK2 channel promotes EMT in PDAC cancer cells

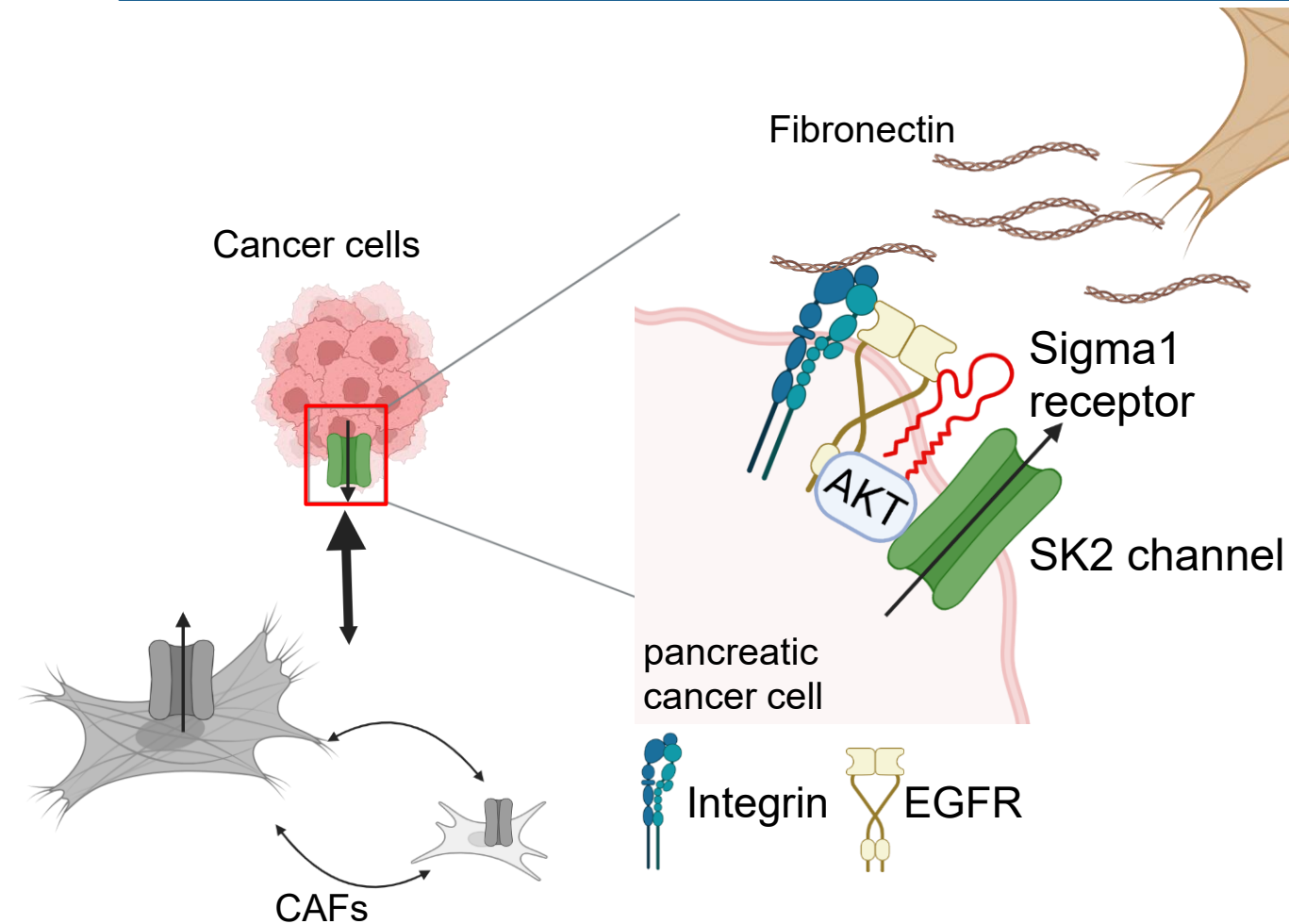


- CAFs secretome (fibronectin) induce the formation of a signalling hub
- Leads to the activation of SK2 channel

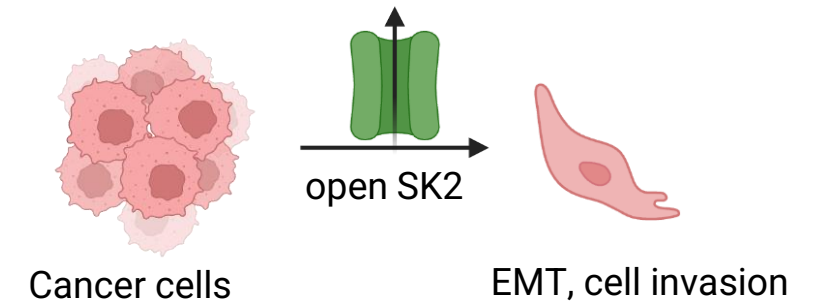


Rapetti Mauss et al, Gut, 2023

SK2 channel promotes EMT in PDAC cancer cells

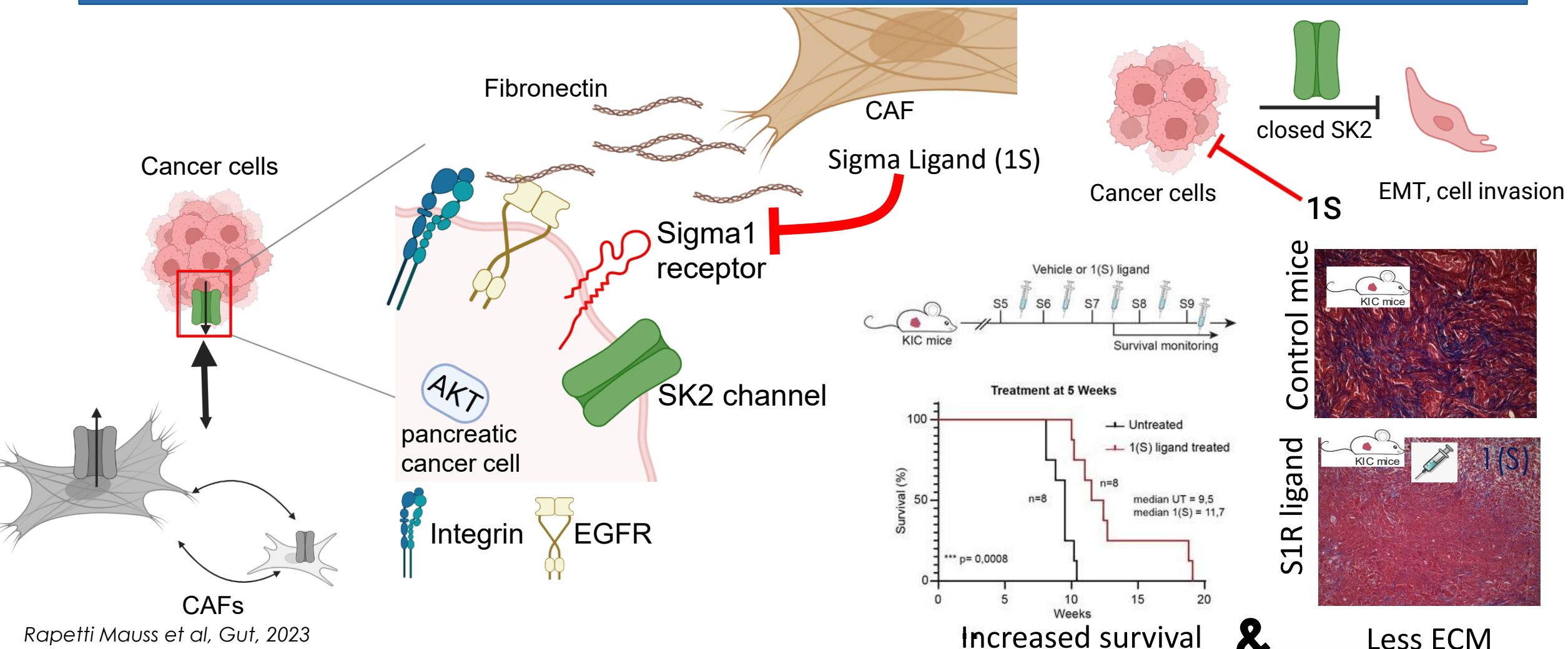


- CAFs secretome (fibronectin) induce the formation of a signalling hub
- Leads to the activation of SK2 channel
- SK2 activation promotes EMT and cell invasion

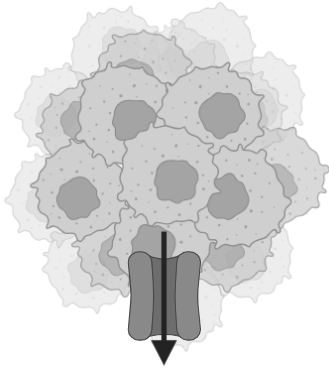


Rapetti Mauss et al, Gut, 2023

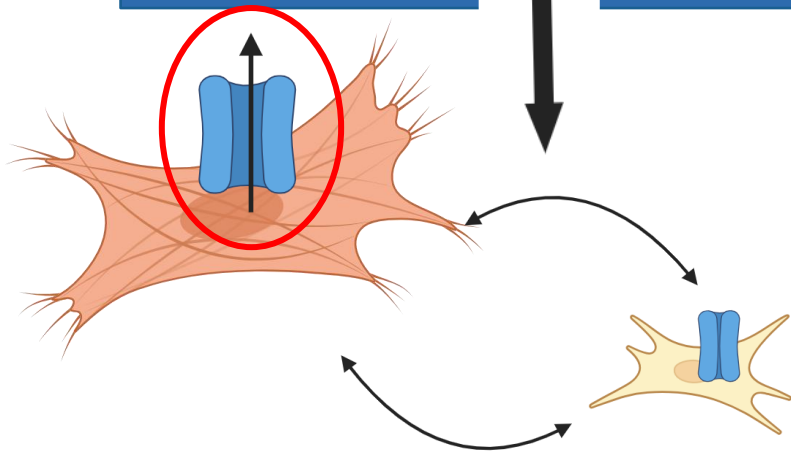
Targeting Sigma1 Receptor (S1R) enhances survival in mice



Cancer cells



Role of S1R and ion channels in CAFs ?



CAFs

Identification of a relevant ion channel : BKCa

Bioinformatic pipeline

Public scRNA seq datas
from patients

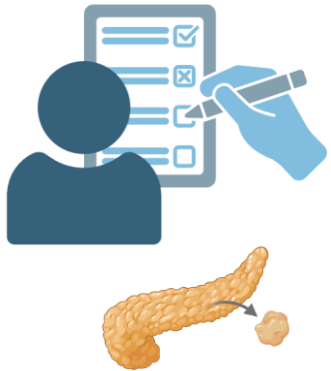


Relevant ion channels in
the stroma



**Identification of BKCa channel
Expressed in CAFs**

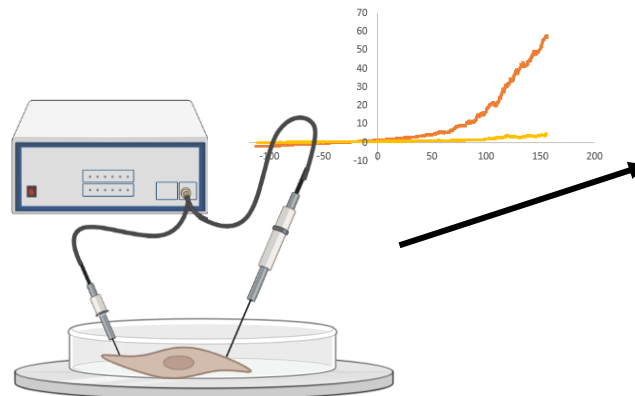
Data : Yousra Missoum



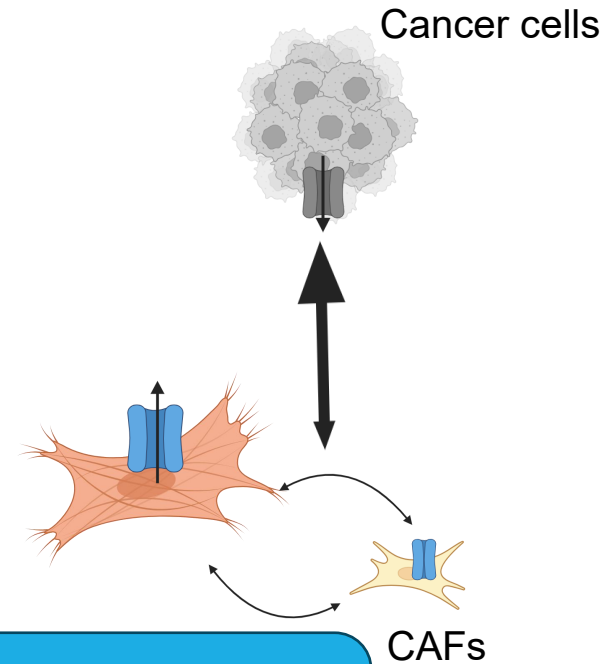
Resection

CAF isolation +
immortalisation

PDAC patient



Patch clamp analysis

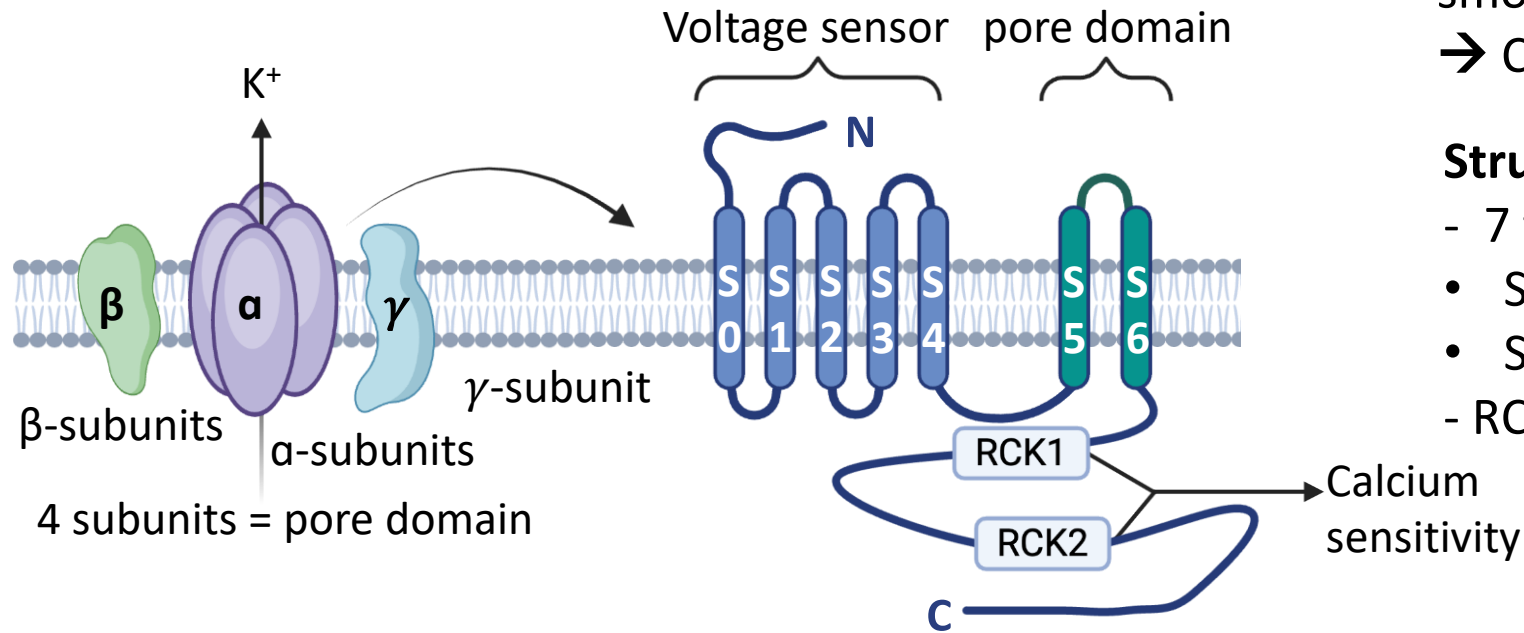


Cancer cells

CAFs

BKCa : a Voltage and calcium-dependant channel

BKCa : big conductance, potassium ion channel



○ α subunits
▲ β/γ subunits

Highly expressed in brain, heart, vascular smooth muscle cells

→ Cardiac rhythmicity, muscle contraction

Structure :

- 7 transmembrane domain :

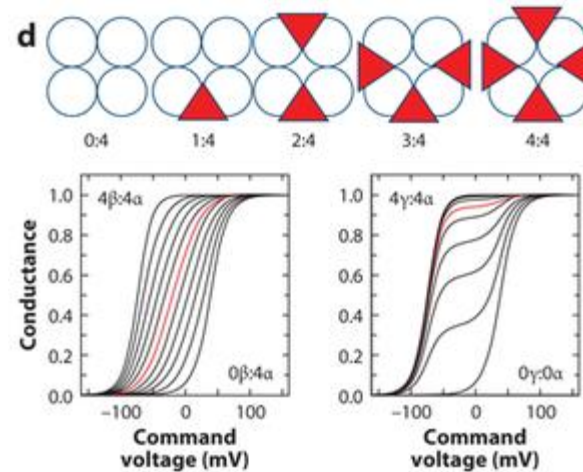
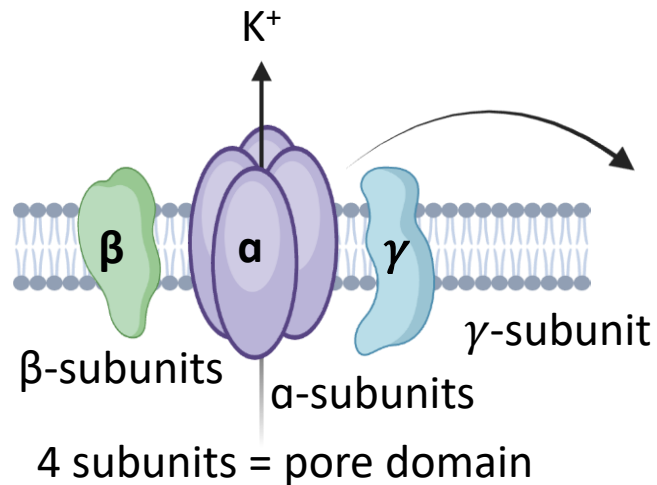
- S1-S4 = voltage sensor
- S5-S6 = channel

- RCK domains = calcium sensitivity

Gonzalez-Perez & J Lingle, 2019

BKCa : a Voltage and calcium-dependant channel

BKCa : big conductance, potassium ion channel



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Gonzalez-Perez & J Lingle, 2019

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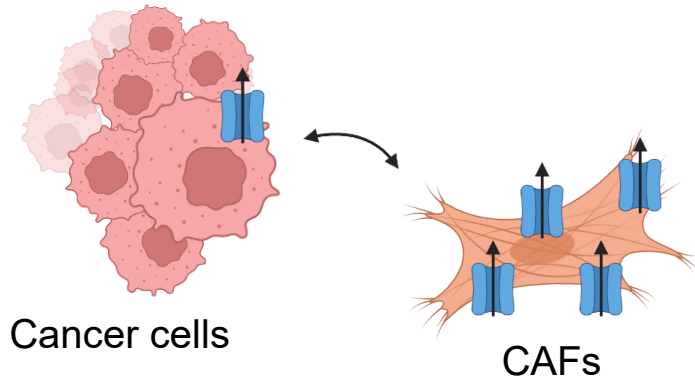
- 7 transmembrane domain :
 - S1-S4 = voltage sensor
 - S5-S6 = channel
- RCK domains = calcium sensitivity

Regulation :

- β and γ - subunits
- Influence voltage-gating, Ca-sensitivity, inhibition (number and stoichiometry)

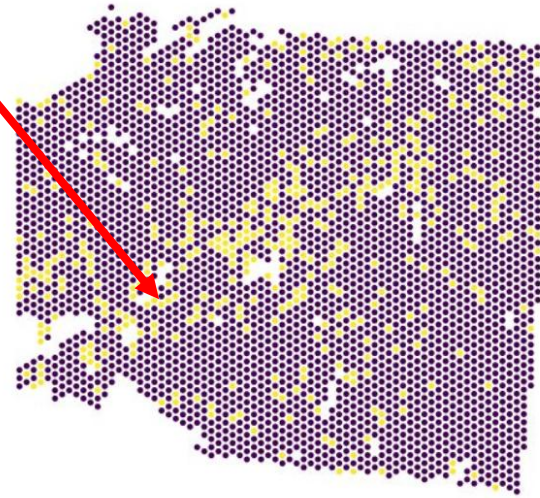
$\beta_1 \rightarrow \beta_4$; $\gamma_1 \rightarrow \gamma_4$

BKCa channel is mainly expressed in the stroma



High BKCa

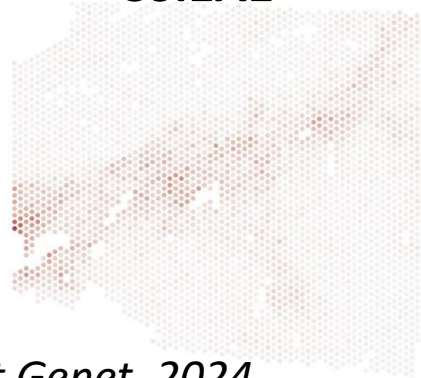
IU_PDA_T1
KCNMA1: C=0.032, Z=7.20, FDR=1.47e-11



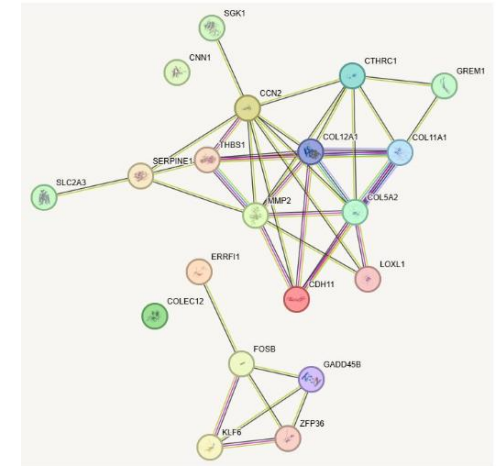
CAFs markers :

Col1A1

Acta2



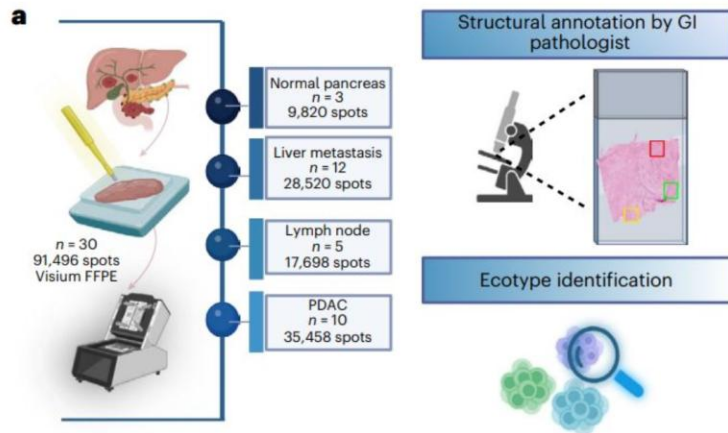
Gene enrichment



- Extracellular matrix
- Collagen containing ECM
- Collagen trimer

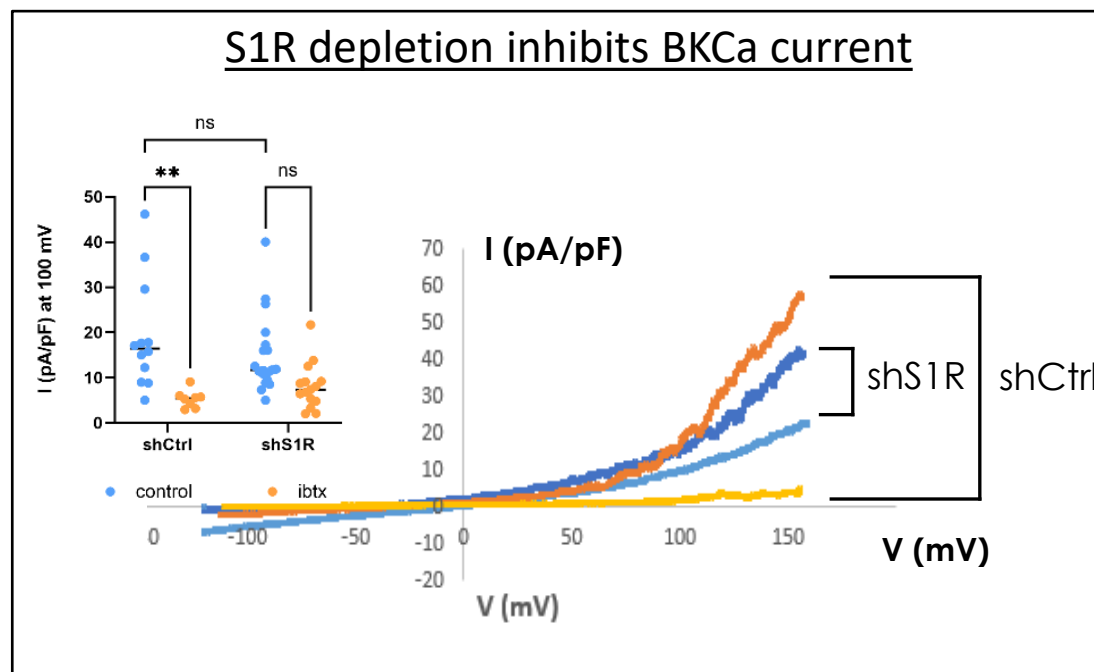
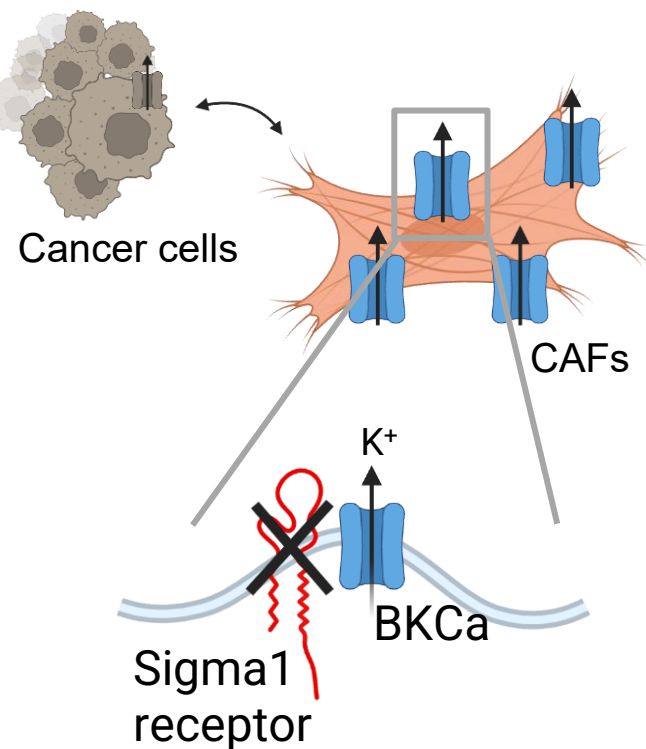
BKCa is specifically associated with CAF markers

Spatial transcriptomic analysis

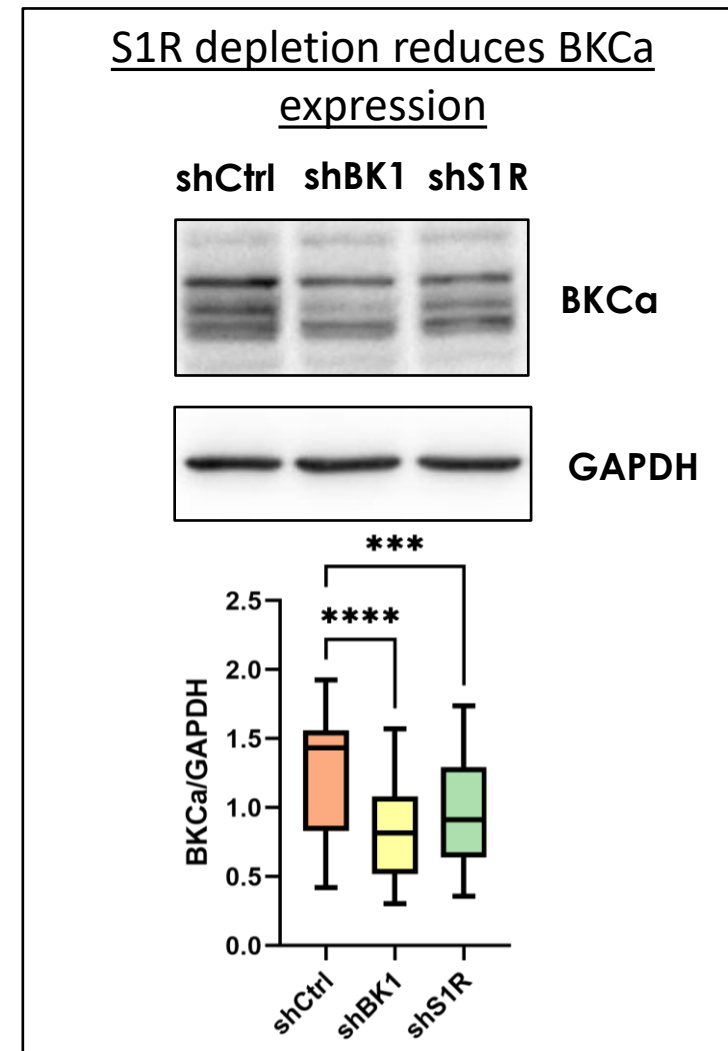


Data : Yousra MISSOUM Khaliq et al., Nat Genet, 2024

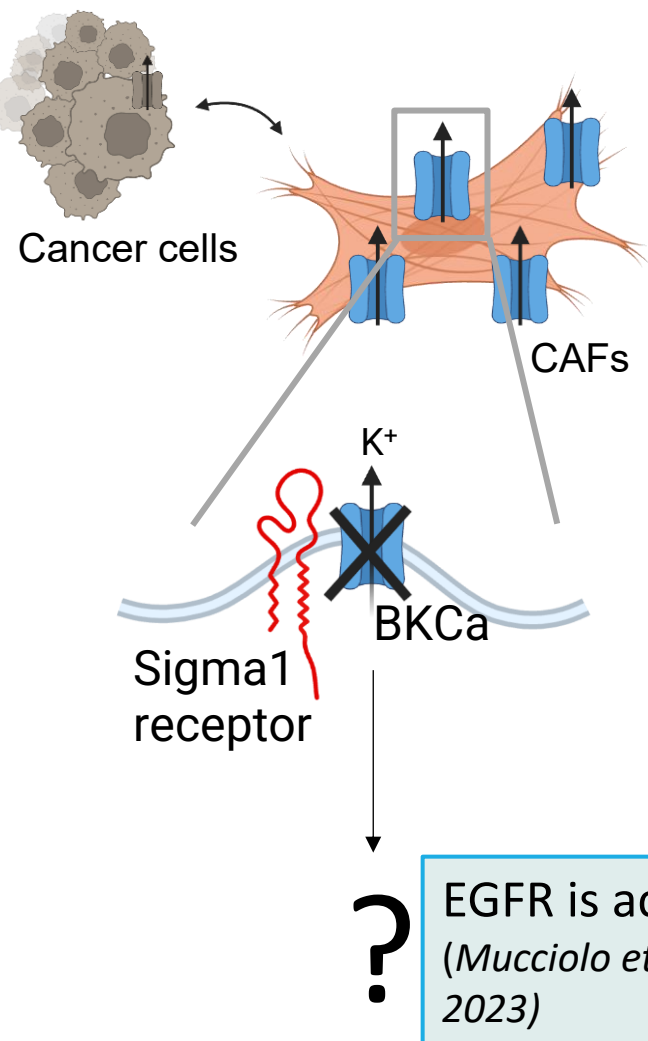
BKCa is regulated by Sigma-1-receptor (S1R) in CAFs



Sigma-1-receptor is a BKCa chaperone of CAFs



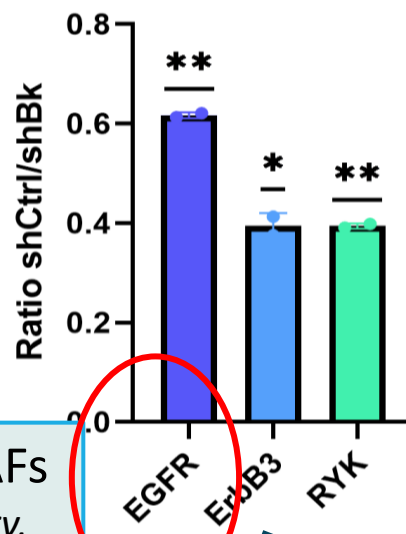
Cell signalling : BKCa is associated to Receptor Tyrosine Kinase (RTK) activation



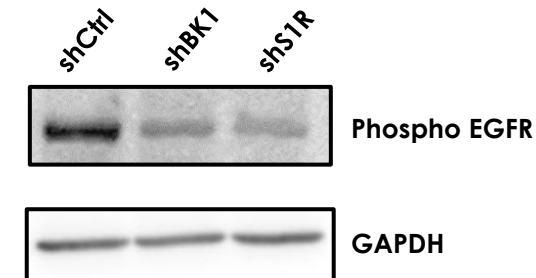
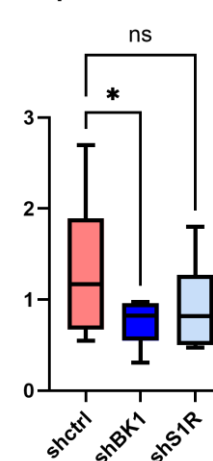
RTK assay (n=2)



Phosphorylation level (shBK1/shCtrl)

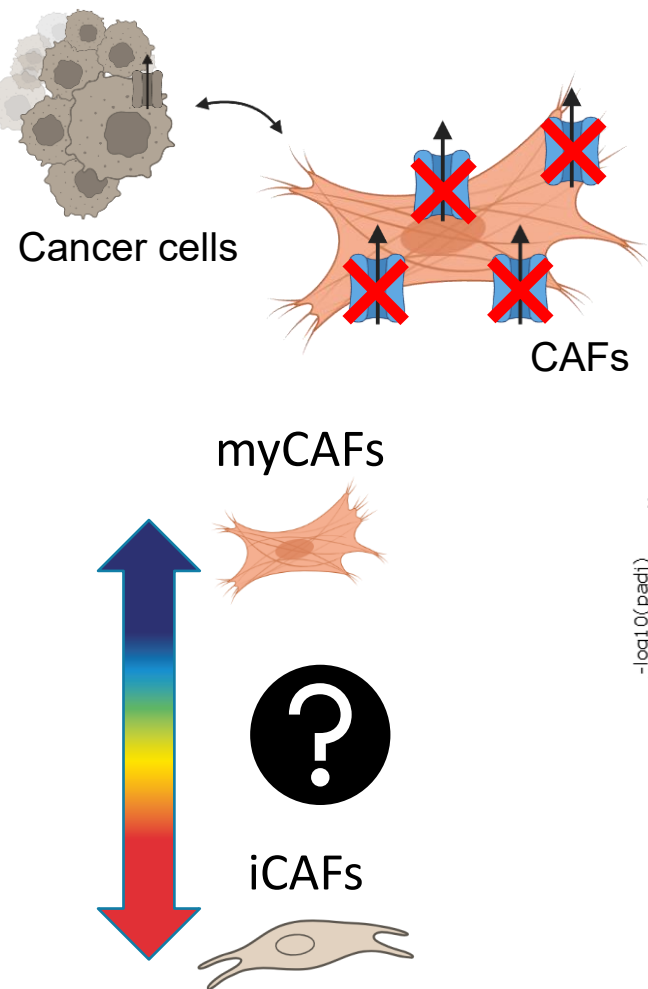


pEGFR/EGFR

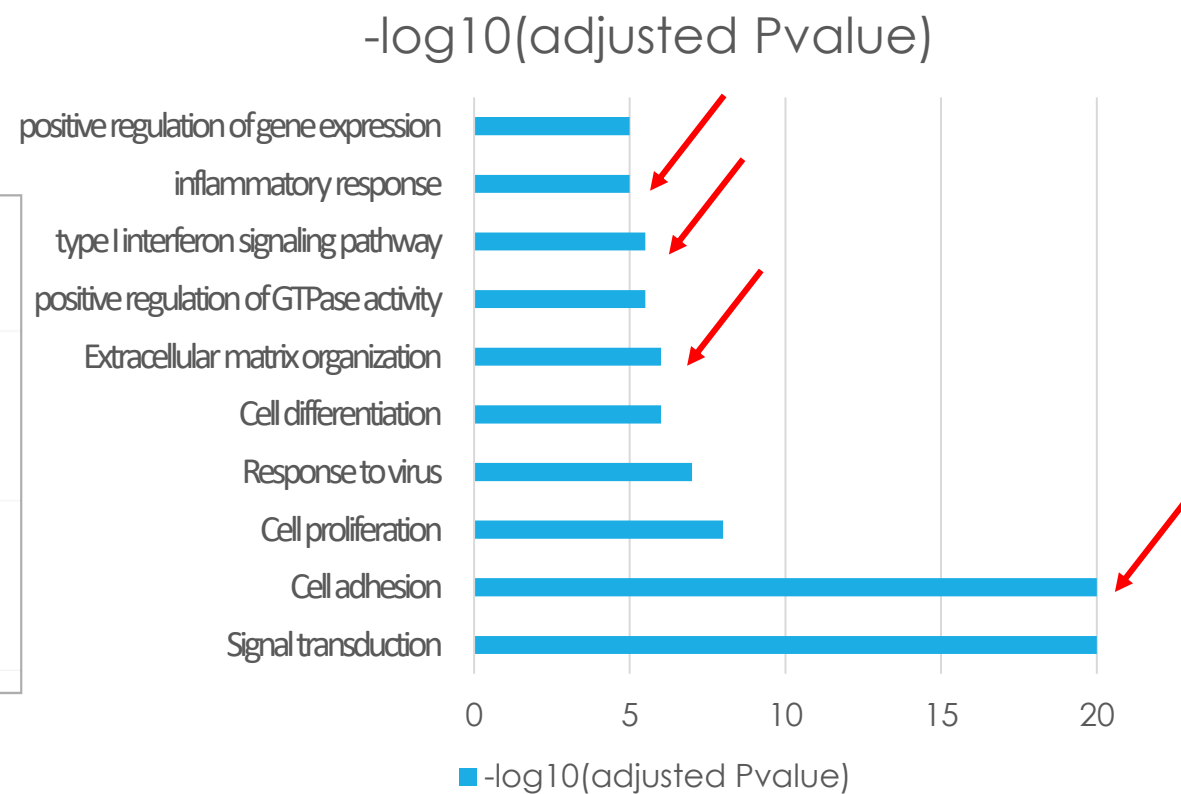
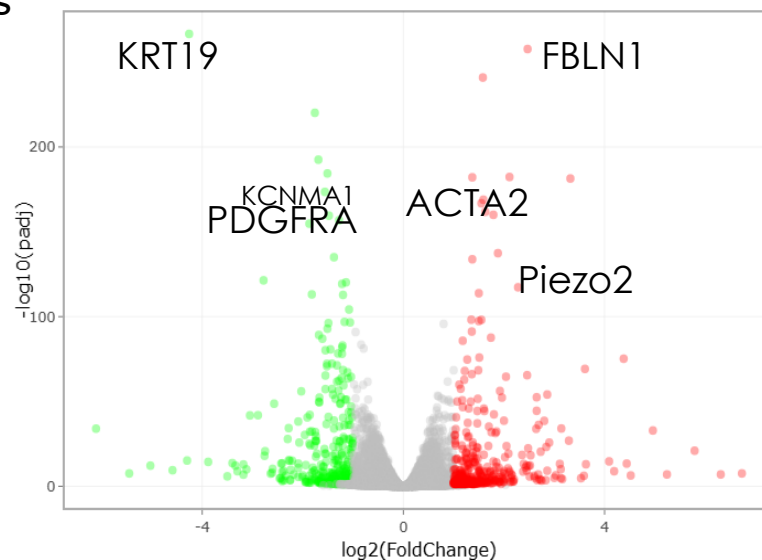


BKCa is implicated in myCAF signaling

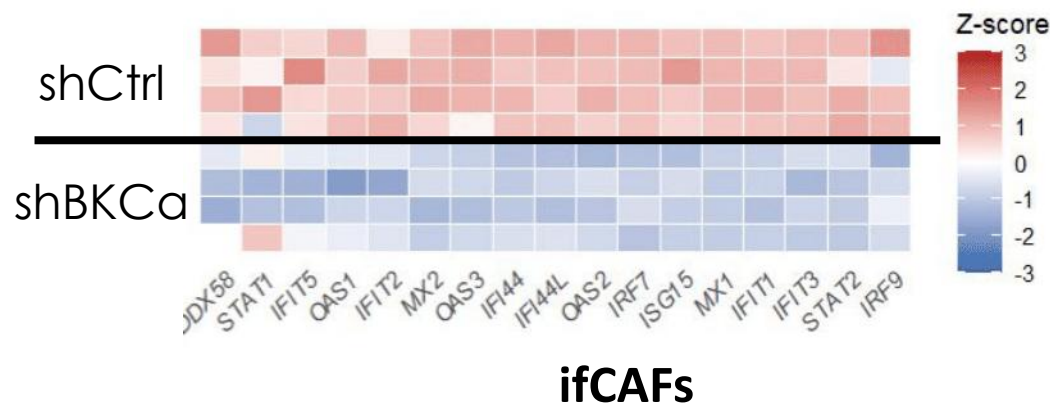
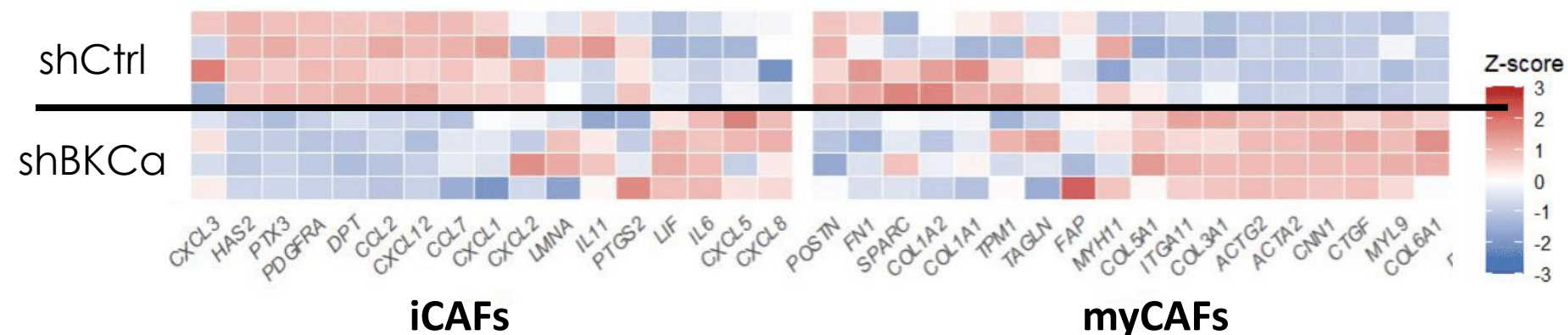
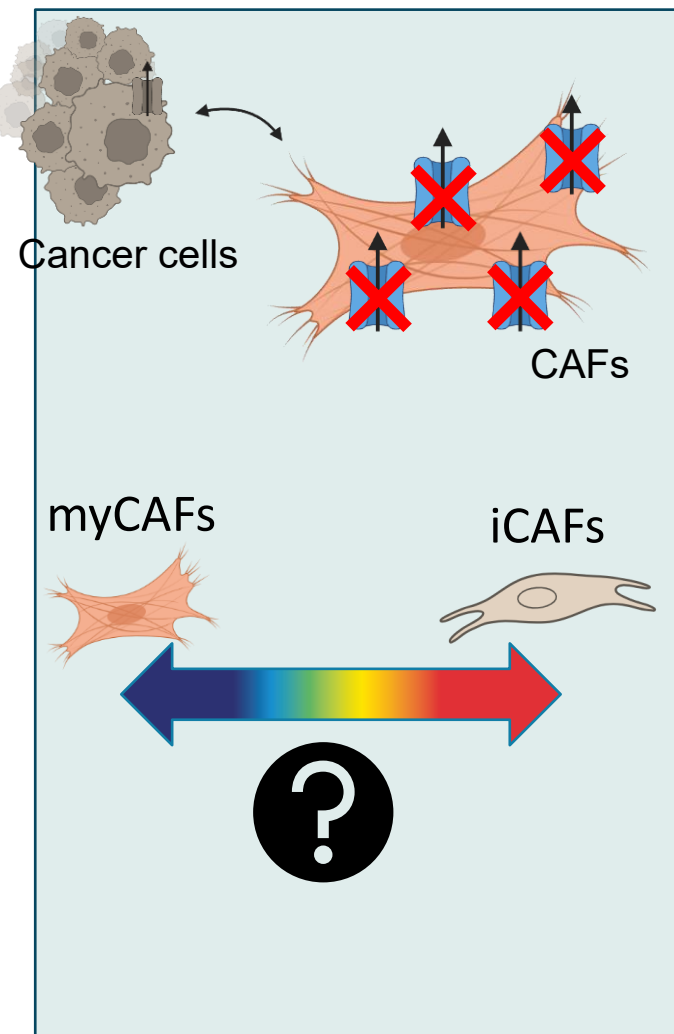
RNA sequencing analysis : BKCa silencing remodels CAF signature



RNA seq analysis : shCtrl vs. shBKCa CAFs



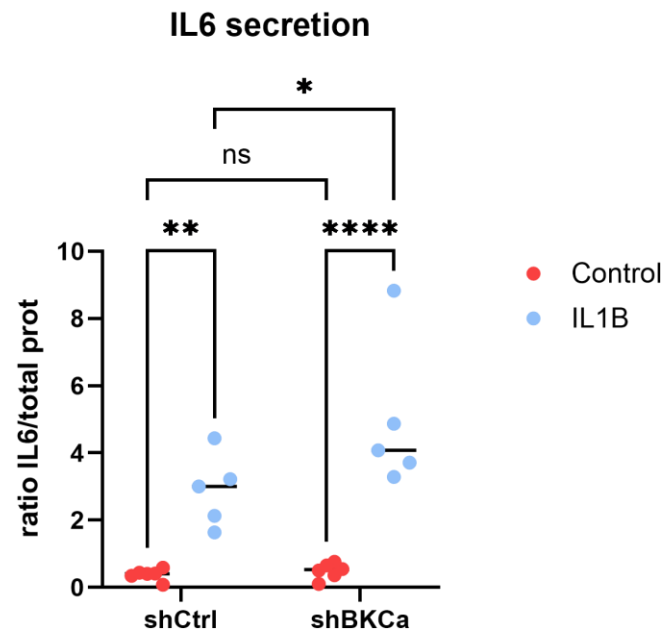
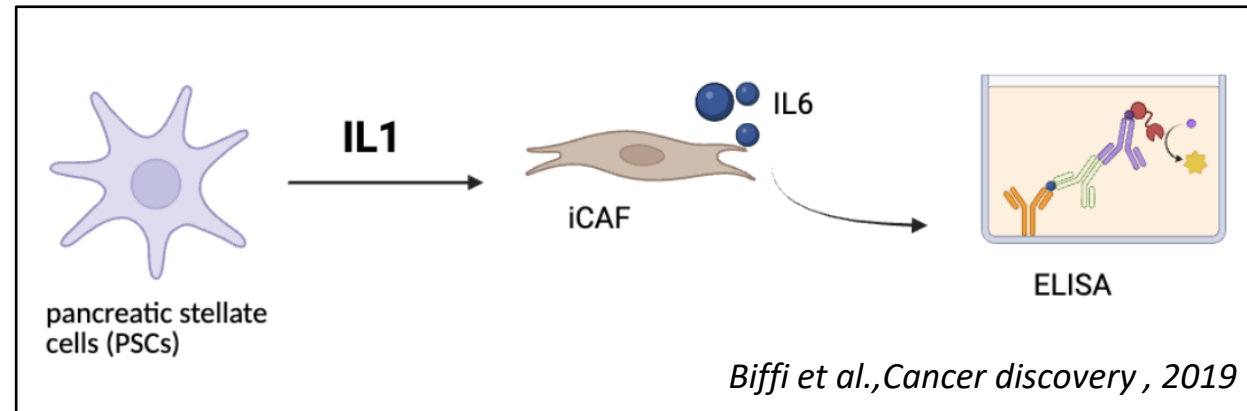
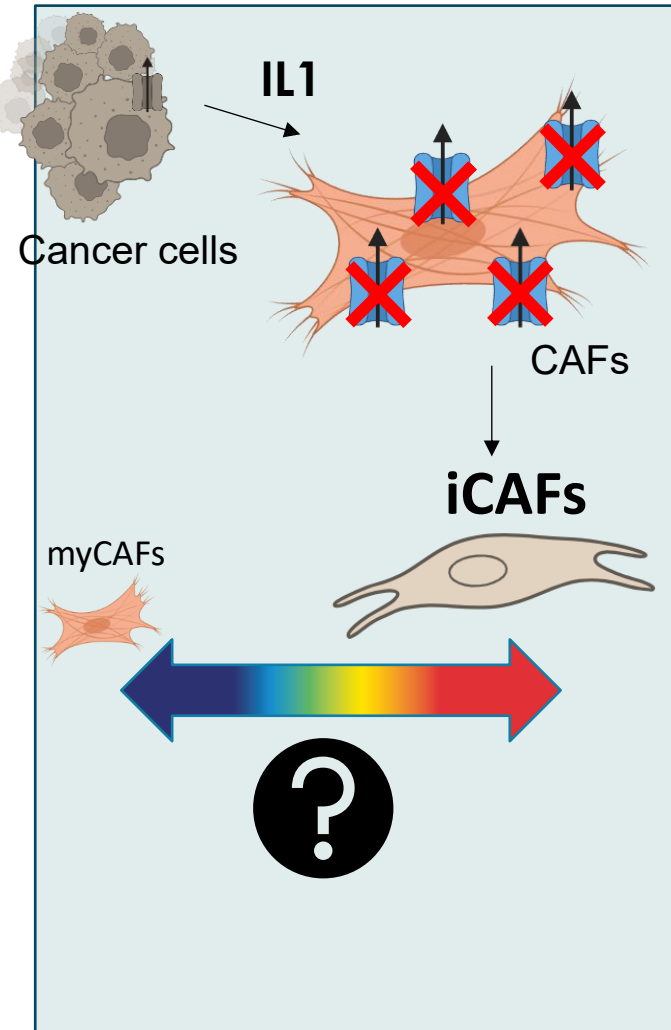
RNA sequencing analysis : BKCa silencing remodels CAF signature



BKCa is a driver of CAFs signature and plasticity

Data : Yousra MISSOUM

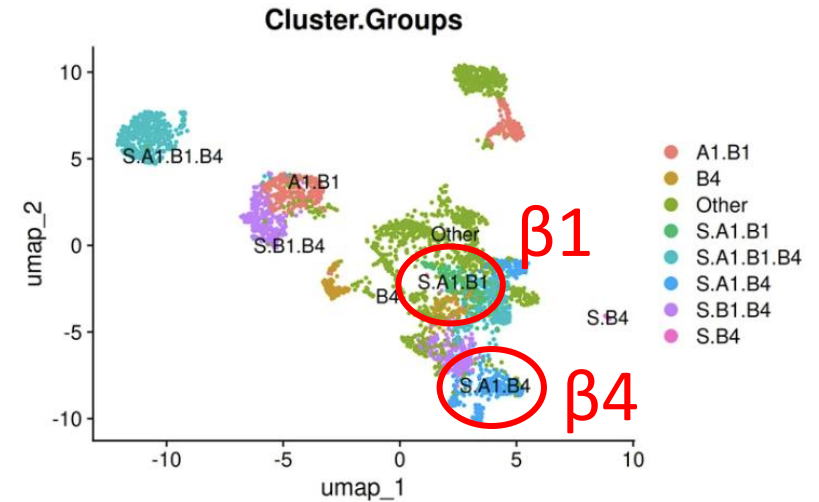
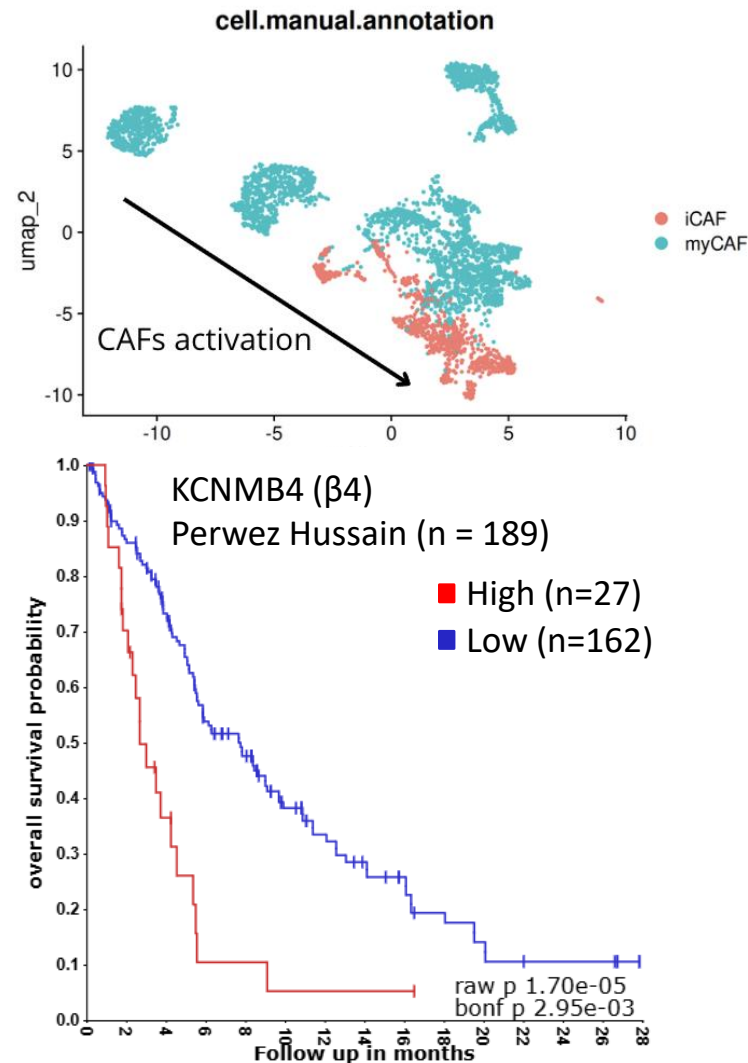
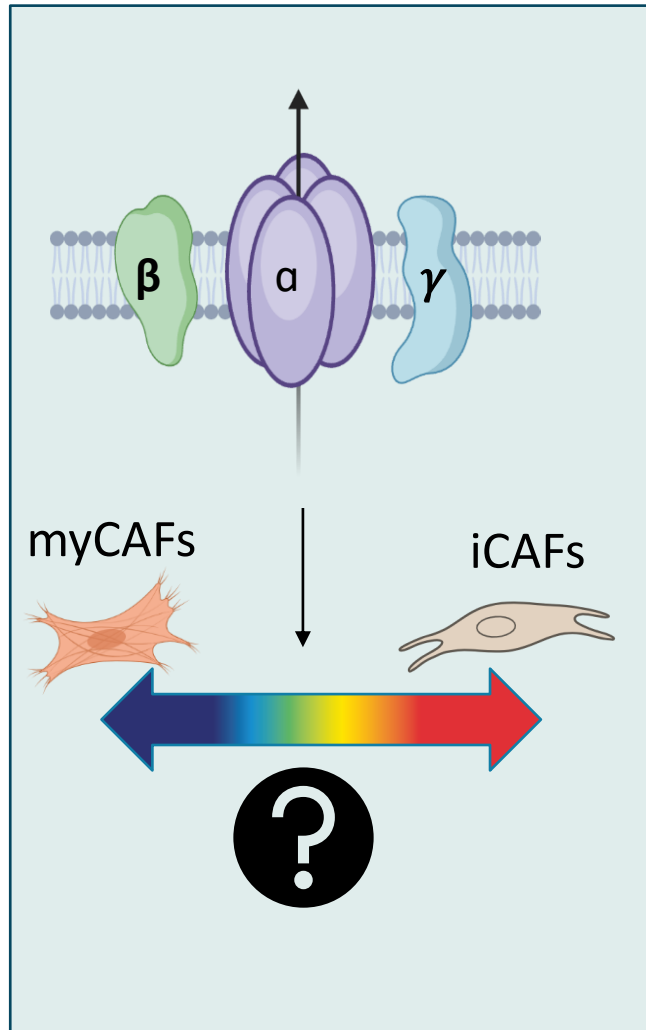
Functional assay : shBKCa CAFs are more sensitive to IL1



=> In response to IL1, shBKCa CAFs produce more IL6

BKCa depletion potentiates iCAFs differentiation

In silico analysis : role of $\beta 1$ and $\beta 4$ subunits



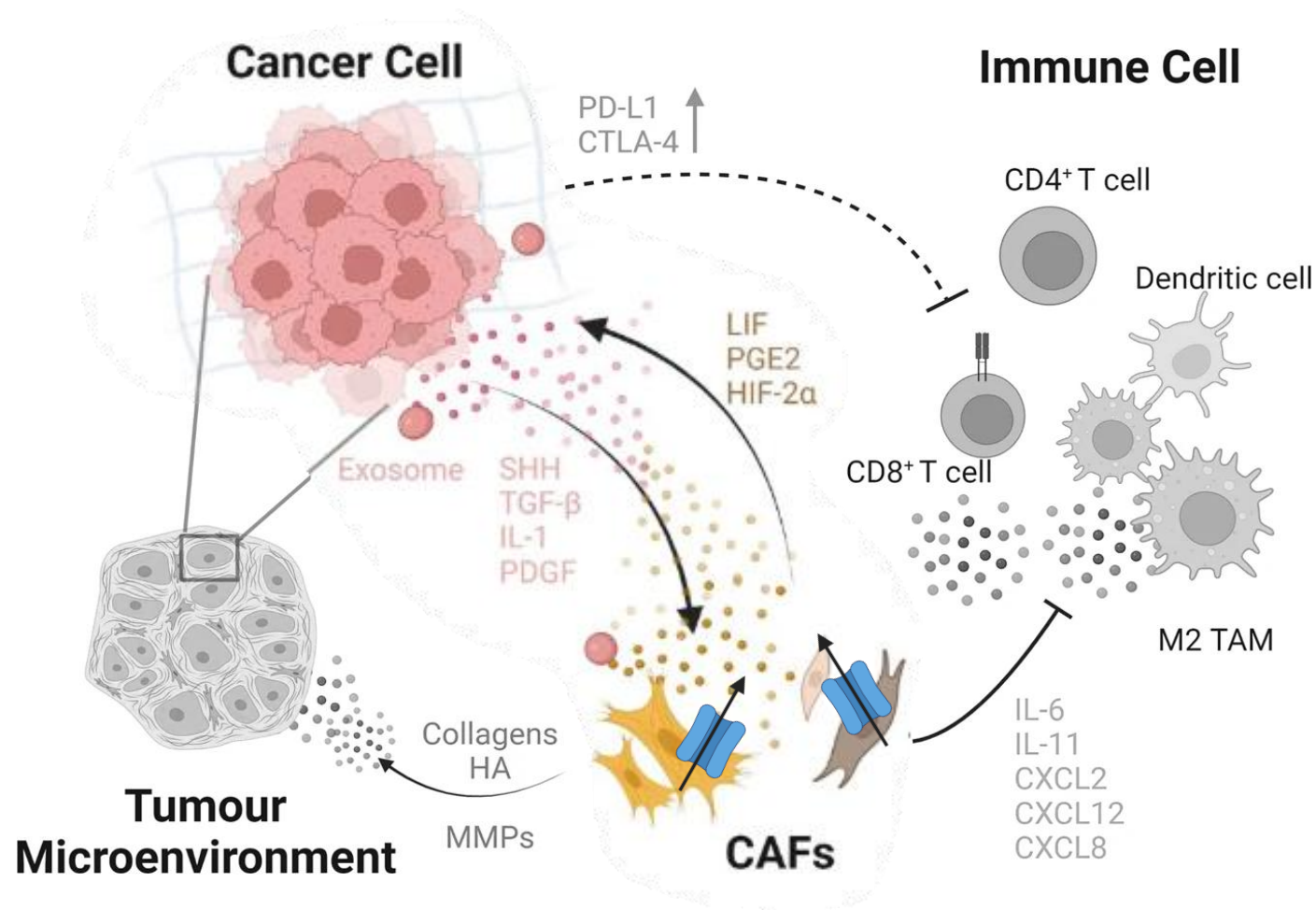
⇒ **$\beta 1$** is associated to **myCAF** phenotype

⇒ **$\beta 4$** is associated to **iCAF** phenotype

⇒ **$\beta 4$** high expression is correlated with lower survival

Data : Claudio MONTENEGRO, CRCM

Conclusion



- BKCa identified as relevant ion channel in the stroma
- Impact on CAF signature
- Impact on CAF sensitivity (IL1)
- Role of $\beta 1$ and $\beta 4$ subunits ?
- Targeting BKCa through Sigma-1-receptor → control CAF plasticity

The team : ion channels and pathologies

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Thank you for your attention



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IBV microscopy platform

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M. Courtillet

IBV cytometry platform

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